# AAE1001 - Introduction to Artificial Intelligence and Data Analytics in Aerospace and Aviation Engineering

## Week 6 – Introduction to GitHub Operations

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# Mutual editing on a document with collaborators remotely?

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| acceptance | 26/3/2021 11:58 AM | File folder | |
| final submission | 2/8/2021 11:38 AM | File folder | |
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| revision 1 | 28/12/2020 5:31 PM | File folder | |
| Artical file (single column).docx | 24/8/2020 3:55 PM | Microsoft Word D... | 4,036 KB |
| Article Processing Charges.pdf | 19/4/2021 9:33 AM | Adobe Acrobat D... | 118 KB |
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| figure.pptx | 22/6/2020 11:11 AM | Microsoft PowerP... | 7,122 KB |
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| Manuscript_two_columns.docx | 13/10/2020 1:45 PM | Microsoft Word D... | 4,030 KB |
| My EndNote Library.enl | 8/1/2021 10:42 AM | EndNote Library | 141 KB |
| References.docx | 14/7/2020 9:16 AM | Microsoft Word D... | 18 KB |

Download files from various channels, such as mail, whatsapp, etc?

Can we edit the documents online ? Google, Microsoft has the solutions!

Can we edit the **“code”** online?

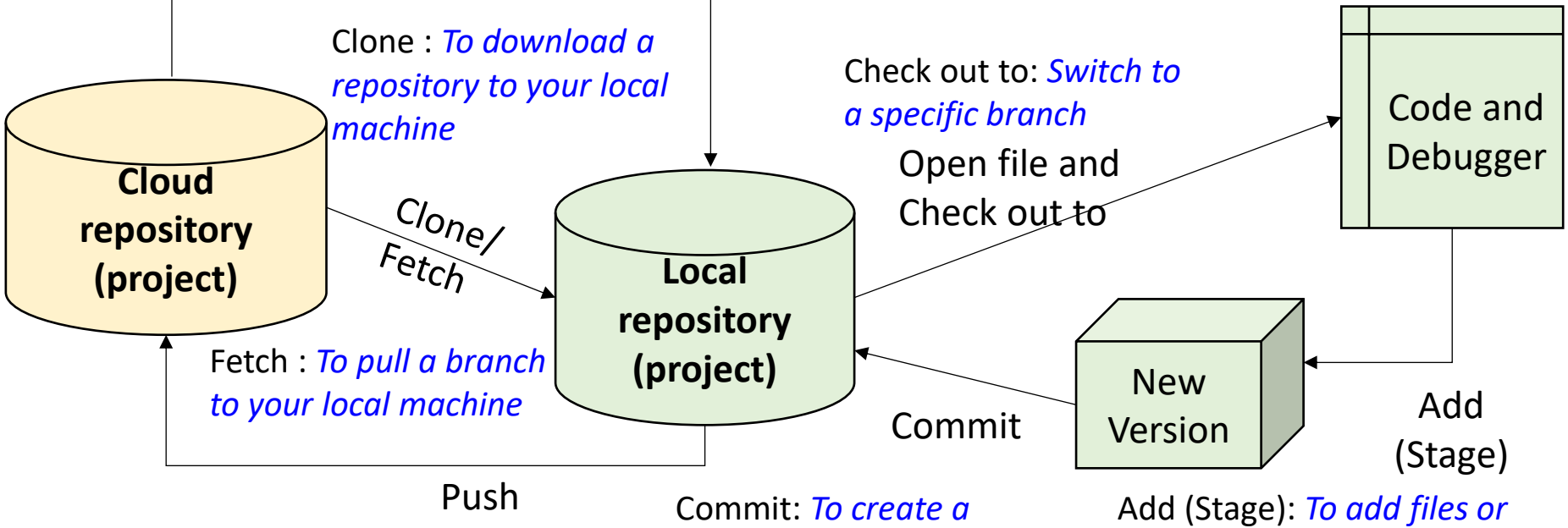
**Github**

# Archive your coding online (in the cloud)

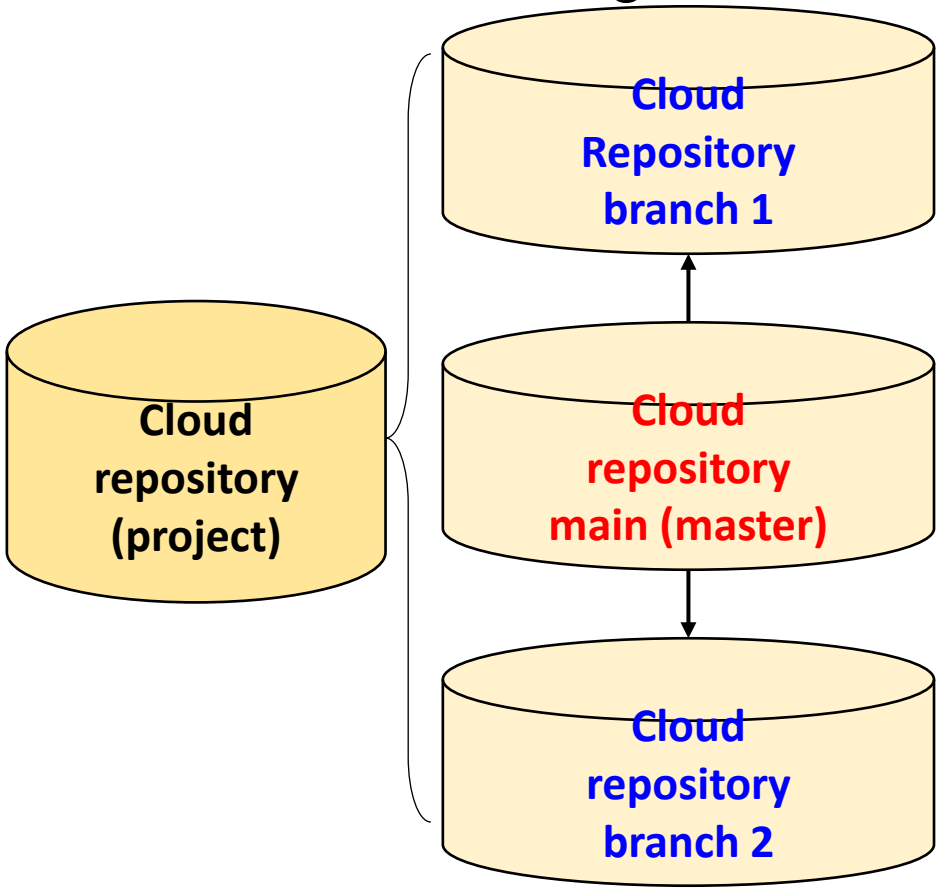
Pull

Pull: *To update local branch with remote, update all remote tracking branches*

Video: VSC-Git Basic Operations

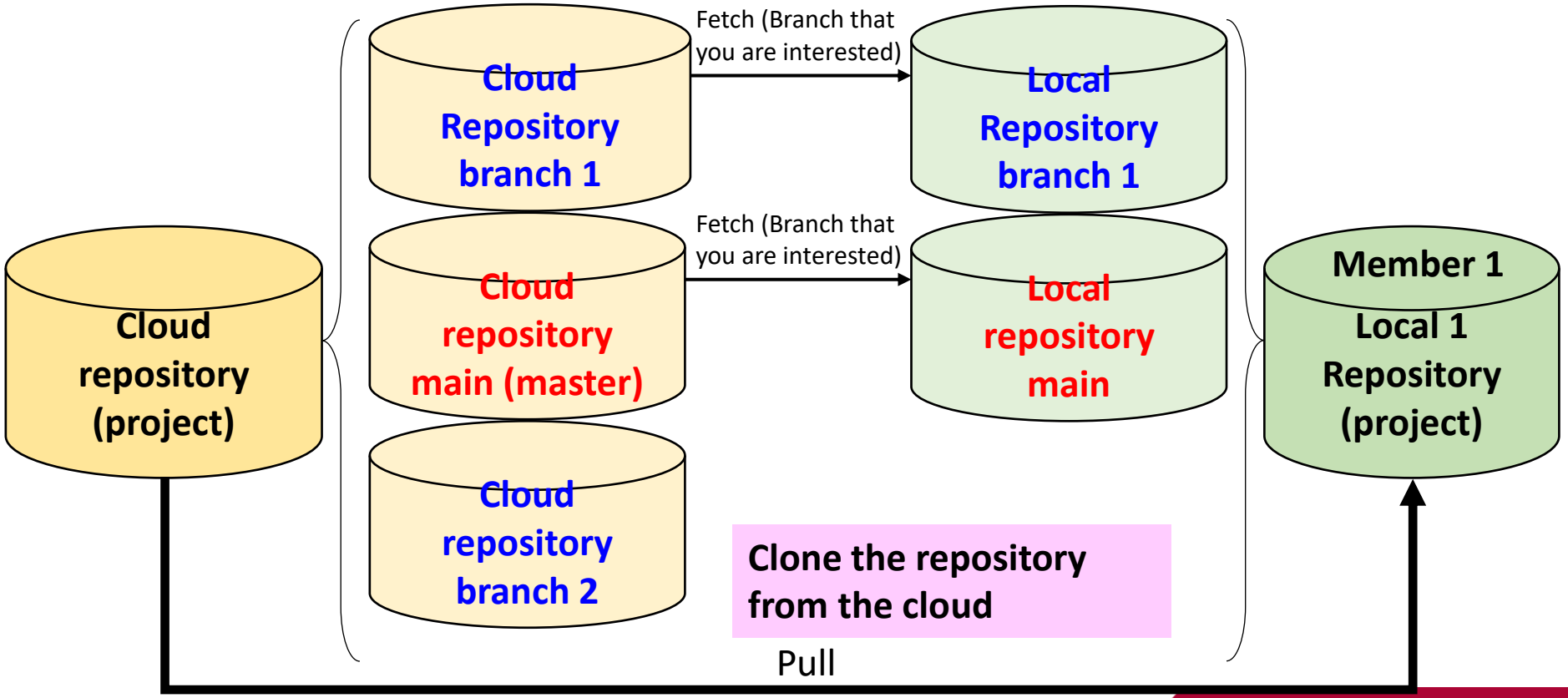


# Collaborative coding online – Generate Branch (Member)

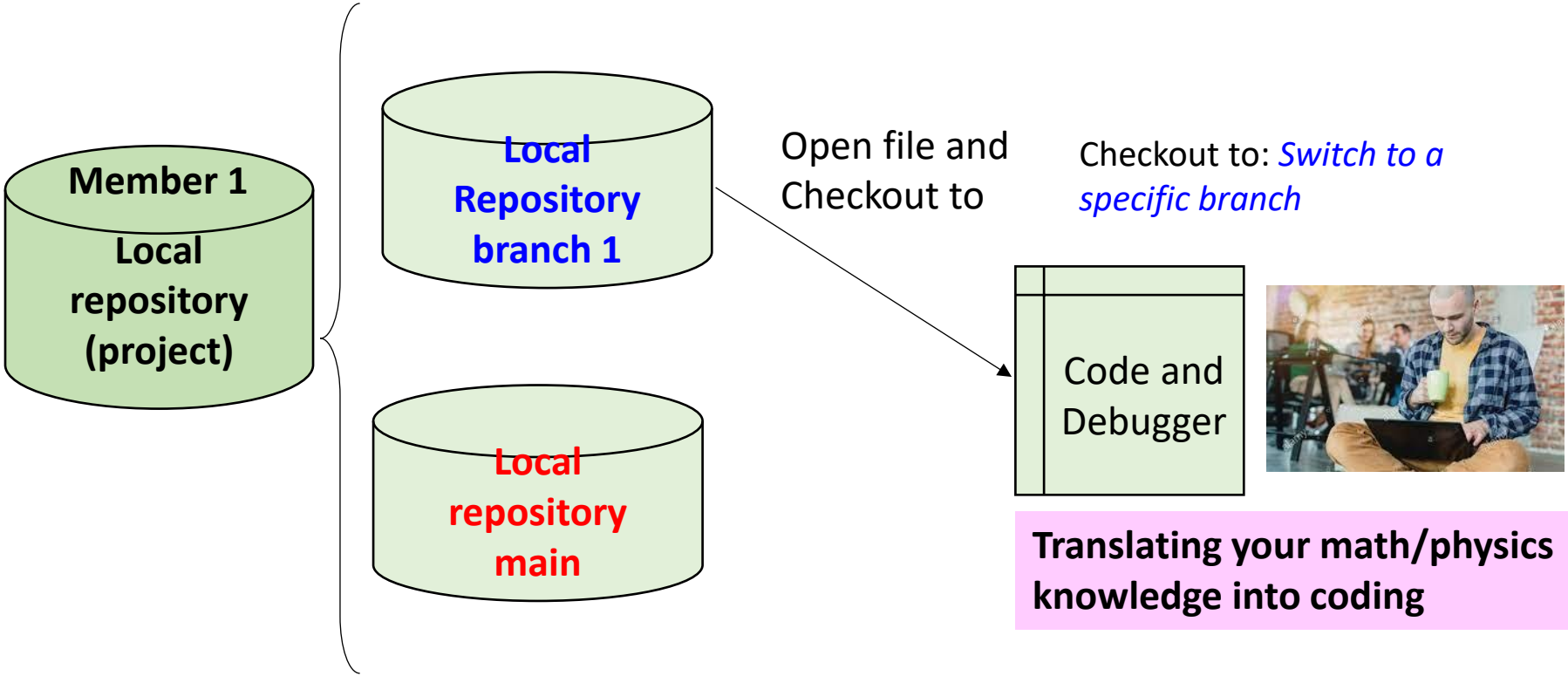



To generate a branch for different member that participant the project, creating a contained area of your repository to develop features, fix bugs and more.

# Collaborative coding online – Fetch/Pull Branch from Cloud

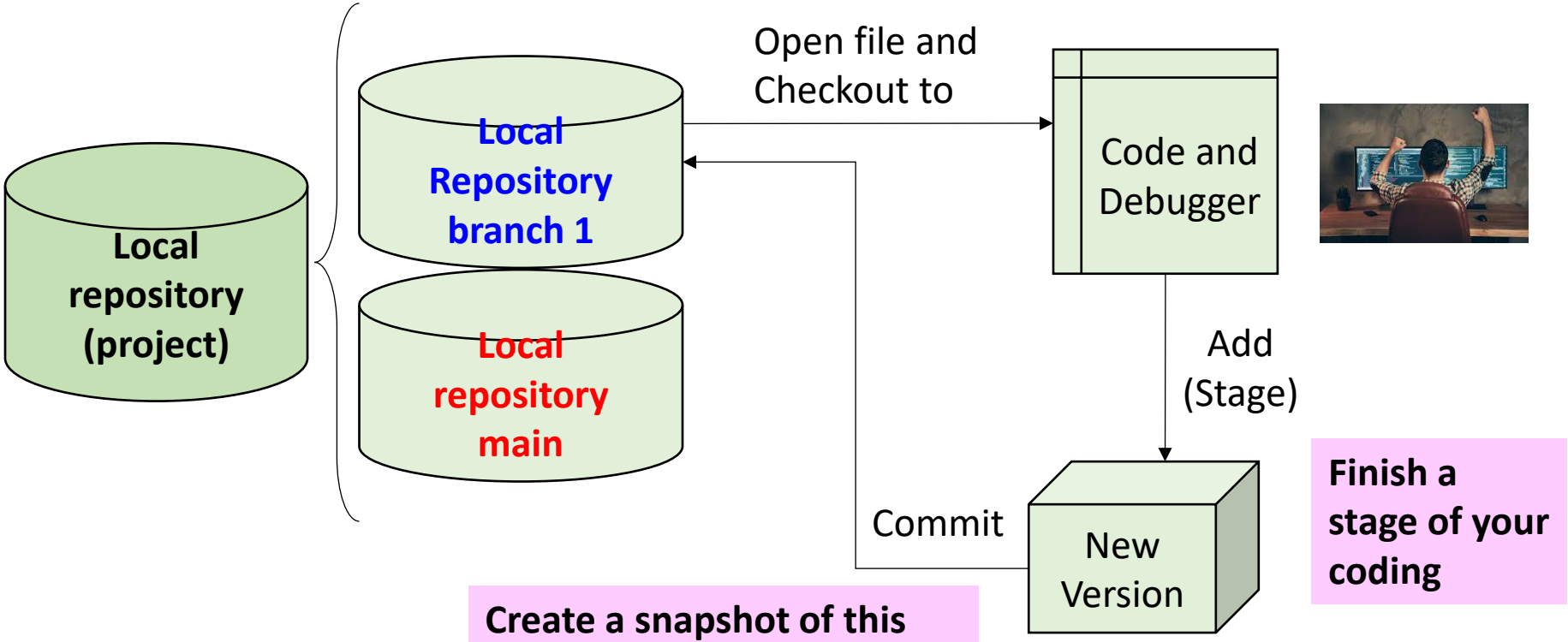


# Collaborative coding online – Edit the code in the local computer




**Translating your math/physics knowledge into coding**

# Collaborative coding online – Edit the code in the local computer



# Concept of collaborative coding online

Web based

Video: VSC-Git Branches

**Ready to update to the cloud branch!**

Open file and Check out to

Code and Debugger

Push

**Cloud Repository branch 1**

**Local Repository branch 1**

File 1 in the local



**Let the group leader knows my work**

Merge by Pull Request

**Cloud repository main**

File 1 in the cloud



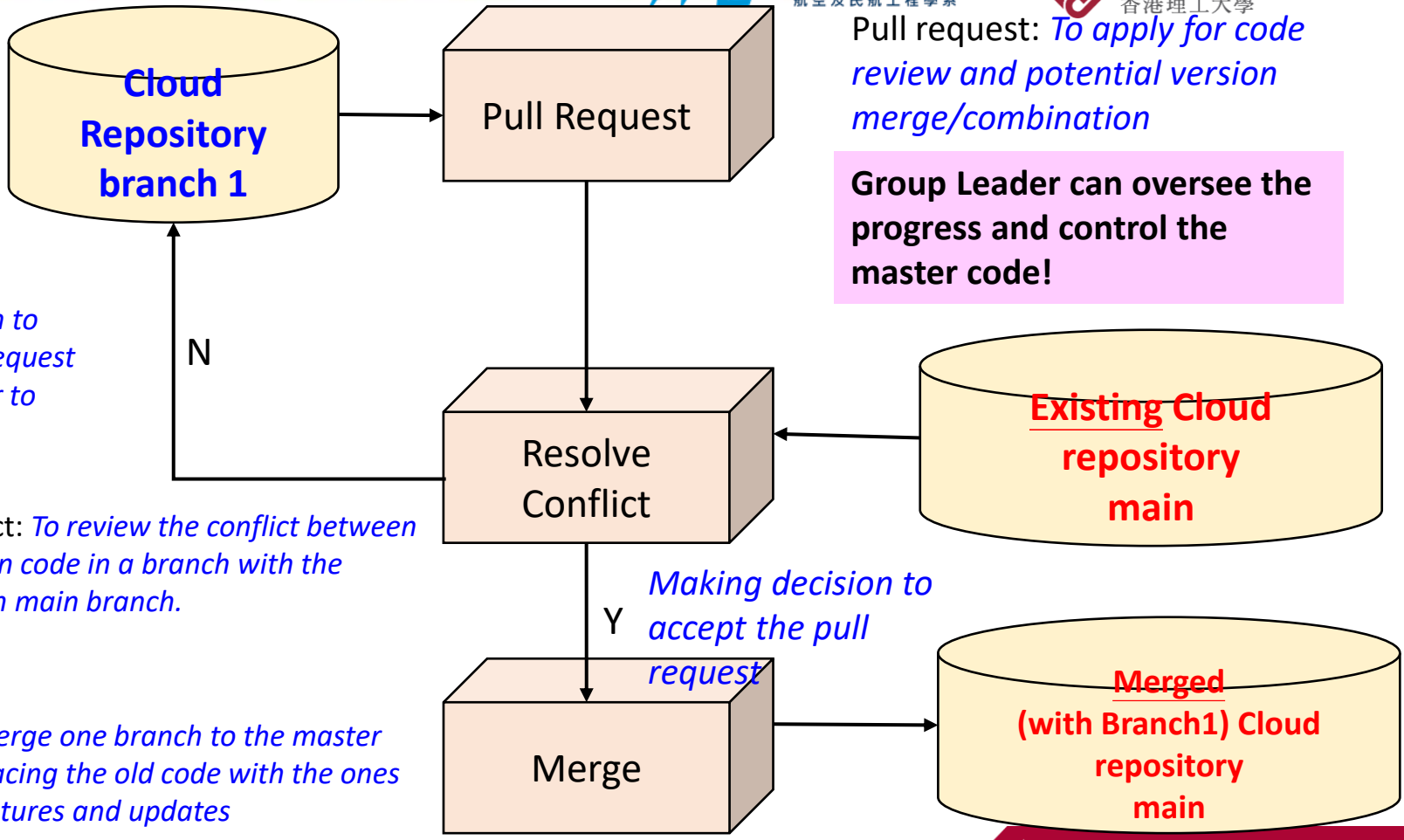
Add (Stage)

Commit

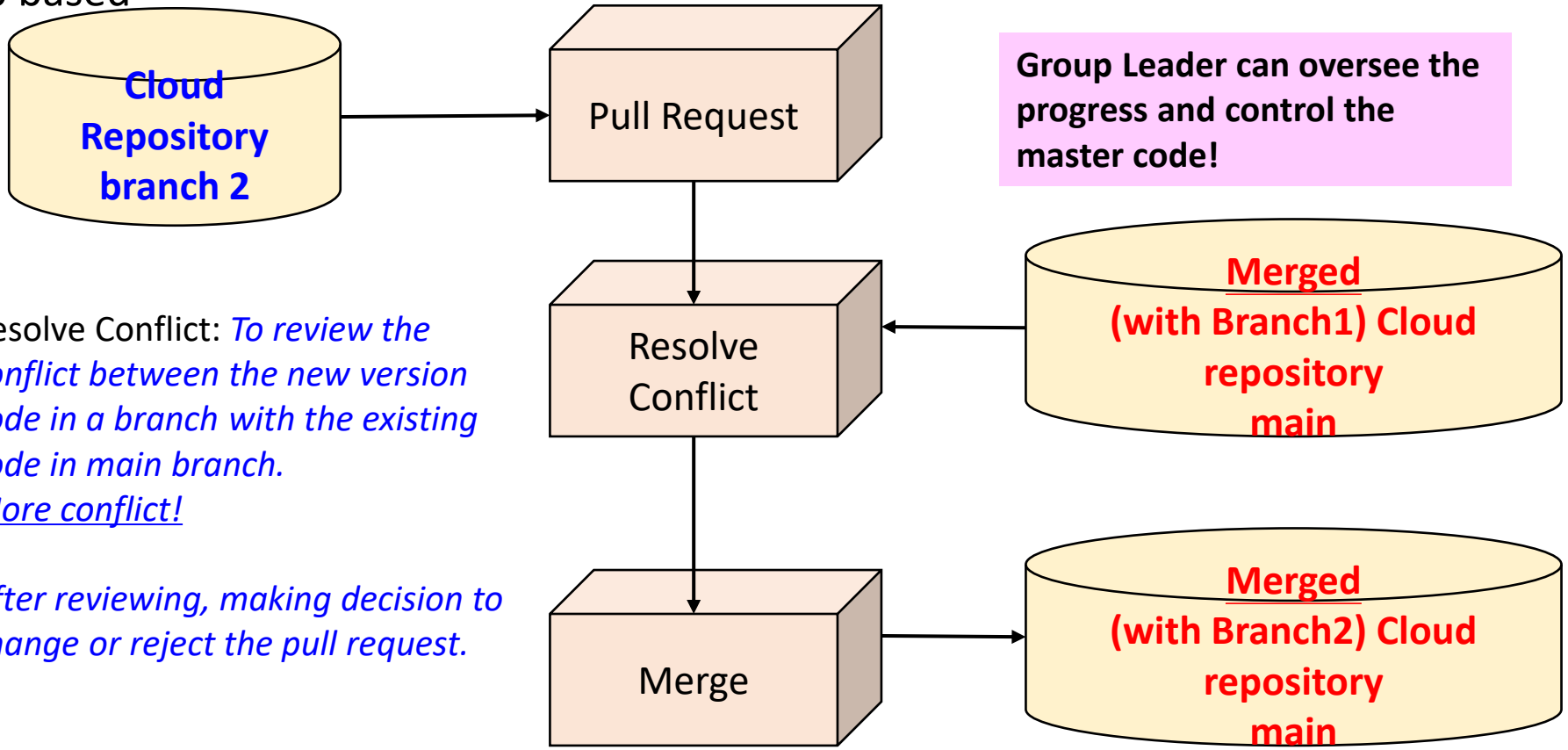
New Version

**If the project leader does not know what you upload to the source code, you will create a lot of confusions!**

Web based


Cloud Repository branch 1
Pull Request
Pull request: To apply for code review and potential version merge/combination
Group Leader can oversee the progress and control the master code!
Making decision to reject the pull request ask the member to revise.
N
Resolve Conflict
Existing Cloud repository main
Resolve Conflict: To review the conflict between the new version code in a branch with the existing code in main branch.
Y
Making decision to accept the pull request
Merge
Merged (with Branch1) Cloud repository main
Merge: To merge one branch to the master branch, replacing the old code with the ones with new features and updates

# Web based




Resolve Conflict: *To review the conflict between the new version code in a branch with the existing code in main branch.*
*More conflict!*

*After reviewing, making decision to change or reject the pull request.*

# GitHub Tutorial Videos (Web Version)

- **Signing Up:** https://www.youtube.com/watch?v=djwgi6YpZWY&ab_channel=POLYUIPNL
- **Group Leader Operations:** https://www.youtube.com/watch?v=xmy9UaXZqAI&ab_channel=POLYUIPNL
- **Group Member Operations:** https://www.youtube.com/watch?v=xVtcexks7EA&ab_channel=POLYUIPNL
- **Creating New Repository + Setup:** https://www.youtube.com/watch?v=fD6x3wborEI&ab_channel=POLYUIPNL
- **Commit & Merge:** https://www.youtube.com/watch?v=34bSuMcdHP4&ab_channel=POLYUIPNL
- **Commit & Merge (rejected):** https://www.youtube.com/watch?v=6zPyvGKE804&ab_channel=POLYUIPNL
- **Searching and Downloading:** https://www.youtube.com/watch?v=5rtr63yCO0c&ab_channel=POLYUIPNL

# GitHub Tutorial Videos (With VS Code)

- **VSC Setup with GitHub:**
  https://www.youtube.com/watch?v=GXbxHbjJQDc&ab_channel=POLYUIPNL
- **VSC Setup with GitHub (Mac):**
  youtube.com/watch?v=iUyr8iG1G8k&ab_channel=POLYUIPNL
- **VSC Git Basic Operation:**
  youtube.com/watch?v=juEEJkgq6fI&ab_channel=POLYUIPNL
- **VSC Git Branches:**
  https://www.youtube.com/watch?v=5qUDlFg5gAM&ab_channel=POLYUIPNL
- **VSC Git Conflicts:**
  https://www.youtube.com/watch?v=4JrCfXhmavg&ab_channel=POLYUIPNL
- **VSC Commits & Pushes with web merges:**
  https://www.youtube.com/watch?v=34bSuMcdHP4&ab_channel=POLYUIPNL
- **VSC Pushing (No web):**
  https://www.youtube.com/watch?v=dSWB5QCUdpE&ab_channel=POLYUIPNL